

MOHAMMAD SHERIFF MEHMOOD

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Senior Data Scientist with 5 years of experience across healthcare, finance, and retail sectors. Expertise in advanced analytics, predictive modeling, and driving data-driven business solutions. Proven ability to deliver impactful insights, optimize processes, and enhance decision-making through cutting-edge techniques and scalable solutions. Skilled in deploying end-to-end projects that achieve measurable results in complex environments.

Skills

Programming	Python, Tensorflow, Numpy, Pandas, NLTK, Spacy,scikit-learn, Docker, LangChain, LangSmith, Django, Flask
Advanced AI and ML	Machine learning, Deep learning, NLP, CNN, Generative AI, MLOps LLMOps, OpenCV, Yolo, ChatBot
Cloud Platform & DB	Google Cloud Platform (GCP), Amazone Web Services(AWS), MongoDB, SQL, Vector DB
Tools	Tableau, Git, Anaconda, JupyterNotebook, Visual Studio, Vertex AI, AWS Sagemaker,
Technical Concepts	CI/CD, Data structure and Algorithms, RAG, Finetunning
Core Competencies	Communication – Secretary of Toastmasters Club; addressed audiences at the PyData Conference.

Professional Experience

Senior Data Scientist - NLP Engineer	LTIMindtree	<i>Pune, INDIA</i>	02/2024 - Present
- Collaborated with a 7-member team to design and implement an AI-powered customer support application. - Lowered customer churn by 40% and boosted quarterly earnings by \$300K through intelligent, context-aware responses. - Engineered DAGs to optimize workflow management, leading to a 40% reduction in processing time. This improvement increased project throughput, enhancing team productivity and speeding up delivery timelines. - Constructed a RAG system for accurate information delivery, decreasing response time by 35% and increasing customer satisfaction. - Built a production-grade LLMOps pipeline using Python sentence-transformers, OpenAI, LangChain, and ChromaDB; optimized deployment and lowered model iteration times by 60%.			
Software Engineer - Data Scientist	Cognizant	<i>Pune, INDIA</i>	06/2021 - 01/2024
- Implemented a targeting model generating \$1M+ in healthcare revenue with 90% accuracy using anomaly detection. - Enhanced accuracy by 5-10% over traditional models with advanced supervised learning and Neural Networks.. - Reduced false alarms in fraud detection by 20% using Logistic Regression and Random Forests, and recived innovation award. - Cut administrative overhead by 40% with an AI-driven prior authorization system. - Created a sophisticated chatbot utilizing Gemini and Vector DB, achieving an impressive 90% accuracy in understanding user queries the tool is now used by over 15 team members for efficient data retrieval. - Engineered a chatbot to extract and process data from PDF, DOCX, and TXT files using advanced language models, improving query accuracy by 40%. - Tech Stack Python, Django, ML/DL frameworks, MongoDB, MLFlow, GCP, LangChain, Faiss, and Git for CI/CD pipeline.			
Programmer Analyst	Cognizant	<i>Chennai, INDIA</i>	12/2019 - 06/2021
- Built data visualization dashboards to track key performance indicators, increasing reporting efficiency by 30%. - Integrated data from over 10 databases, APIs, and external sources, improving data consistency by 25%. - Supported the development and training of machine learning models, increasing prediction accuracy by 20% and saving \$200K in costs under senior guidance.			

Education

Master of Technology in Data Science & Engineering

BITS Pilani Pilani, India
09/2021 - 10/2023

Bachelor of Engineering in Computer Science & Engineering

Oriental Institute of Science & Technology Bhopal, India
08/2015 - 06/2019

Certifications and Awards

- **First Prize*** : Secured top prize at the Trizetto Hackathon for a patient data integration approach, earning 95% positive feedback.
- **OpenAI Whisper Hackathon:** Developing an app to assist individuals with hearing disabilities and improve their quality of life.
- **GCP Professional ML Engineer:** Certified in Google Cloud Platform (02/2023).
- **Certified MLOps Developer:** Dataiku certification for ML model operations.
- **IBM Certification:** IBM professional data science certification.

Research Projects

Accident/Fall Detection

- Worked on real-time human fall detection system using Python, YOLO, and **OpenCV** with CUDA toolkit, achieving **95%** accuracy in fall detection.
- Designed a smart camera surveillance system for highway accident detection using CNN and YOLO. Automated ambulance dispatch and decreased emergency response time by 40% using Python and its SMTP library for alert email generation.

Disease Classification

- Developed a mobile app for classifying potato diseases with **92%** accuracy in detecting **Early** and **Late Blight**
- Quantized the CNN model with TensorFlow Lite, reducing model size by **80%** and improving inference speed by 50%.
- Utilized Google Cloud Functions for the system, achieving a **25%** reduction in cloud costs, and integrated TensorFlow Serving with FastAPI for real-time predictions on ReactJS and React Native interfaces.
- **Technologies: TensorFlow, CNN, TF Lite, Data Augmentation, TF Serving, FastAPI, ReactJS, React Native, Google Cloud Functions.**

Abstractive Text Summarization

- Applied BERT and NLP toolkits to summarize chat context, improving agent efficiency by **35%** and enhancing customer service response times by **25%**.
- Improved agent efficiency by providing quick access to previous chat history for better customer support.

Visa Approval Forecasting for US Immigration

- Applied production-grade code to predict structure data. Provide database connectivity with **MongoDB**. EDA and feature engineering
- Explored various ML models to achieve optimal results through gridSearchCV and hyperparameter tuning, and containerized the solution with Docker for AWS integration.

Automated Topic Modeling and Deployment on AWS for Business Insights

- Established an end-to-end machine learning pipeline for topic modeling on AWS, cutting processing time by **40%**.
- Dockerized and deployed a Python-based LDA topic modeling application on AWS ECS, cutting deployment time by 60% through a blue/green deployment strategy.
- Automated CI/CD pipelines, reducing manual effort by **70%** using **AWS CodeCommit, CodeBuild, CodeDeploy, MLOPS and CodePipeline**, while ensuring high availability with **EC2 and ECS** load balancing.Used MLflow for model version controlling.

Automated News Article Categorization

- Developed a high-accuracy text classification model to categorize news articles into four categories (World, Sports, Business, Science/Technology), automating the categorization process.
- Implemented modular Python code for data processing and modeling, achieving streamlined workflows.
- Deployed on GCP, reducing data processing time by 40% and improving scalability.